SAT Encodings

Full Adder

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Addition in Propositional Logic

Decision problem

Given A and B (represented in binary), find D

Such that $A + B = D$

Variables

Inputs:

$A_0, \ldots, A_n$

$B_0, \ldots, B_n$

Output:

$D_0, \ldots, D_n$

Carry:

$C_0, C_1, \ldots, C_n$

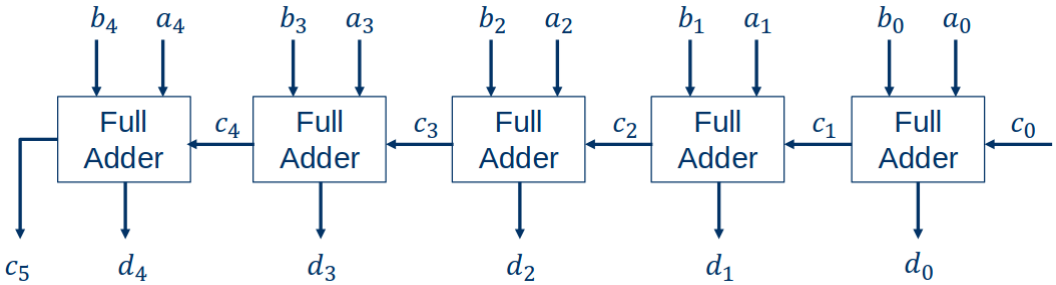
n-bit Binary Adder using Full Adders

Chain of Full Adders

Each bit position uses one Full Adder

Inputs: A_i, B_i, C_i

Outputs: D_i, C_{i+1}



Full Adder Encoding

Sum bit

$$D_i = A_i \oplus B_i \oplus C_i$$

Implementation

$$D_i \iff (A_i \wedge \neg B_i \wedge \neg C_i) \ \vee \ (\neg A_i \wedge B_i \wedge \neg C_i) \vee$$
$$(\neg A_i \wedge \neg B_i \wedge C_i) \ \vee \ (A_i \wedge B_i \wedge C_i)$$

$$C_{i+1} \iff (A_i \wedge B_i) \ \vee \ (A_i \wedge C_i) \ \vee \ (B_i \wedge C_i)$$

Other Constraints

Boundary Conditions

$$C_0 = \text{False} \qquad A_n = \text{False}, \qquad B_n = \text{False}$$

Input Constraints

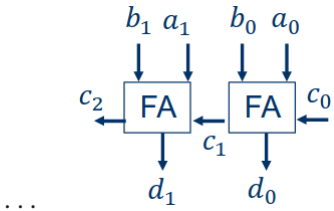
$A[i] = \text{True} \quad \forall$ 1s in binary representation
$A[i] = \text{False} \quad \forall$ 0s in binary representation
$B[i] = \text{True} \quad \forall$ 1s in binary representation
$B[i] = \text{False} \quad \forall$ 0s in binary representation

AtMostOne via Full Adder Encoding

Idea

- Use Full Adders to sum Boolean variables
- Restrict the sum to be less or equal to one

AtMostOne$([x_1, x_2, \ldots x_n])$ Enforce: $\sum_{i=1}^{n} x_i \leq 1$



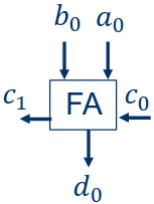
AtMostOne with 2 Variables

AtMostOne$([x_1, x_2])$ Enforce: $x_1 + x_2 \leq 1$

Encoding using One Full Adder

Full Adder:

$$A_0 = x_1$$
$$B_0 = x_2$$



Constraints: $\neg C_0 \wedge \neg C_1$

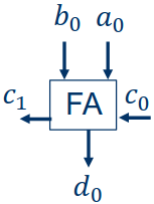
AtMostOne with 3 Variables

AtMostOne$([x_1, x_2, x_3])$ Enforce: $x_1 + x_2 + x_3 \leq 1$

Encoding using One Full Adder

Full Adder:

$$A_0 = x_1$$
$$B_0 = x_2$$
$$C_0 = x_3$$



Constraints: $\neg C_1$

AtMostOne with 4 Variables

AtMostOne$([x_1, x_2, x_3, x_4])$ Enforce: $x_1 + x_2 + x_3 + x_4 \leq 1$

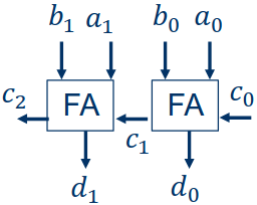
Encoding using two Full Adders

First FA:

$$A_0 = x_1$$
$$B_0 = x_2$$

Second FA:

$$A_1 = x_3$$
$$B_1 = x_4$$



Constraints: $\neg C_0 \wedge \neg C_1 \wedge \neg C_2 \wedge (\neg D_2 \vee \neg D_0)$

AtMostOne with 5 Variables

AtMostOne$([x_1, x_2, x_3, x_4, x_5])$ Enforce: $x_1 + x_2 + x_3 + x_4 + x_5 \leq 1$

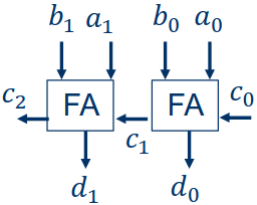
Encoding using two Full Adders

First FA:

$$A_0 = x_1$$
$$B_0 = x_2$$
$$C_0 = x_3$$

Second FA:

$$A_1 = x_4$$
$$B_1 = x_5$$



Constraints: $\neg C_1 \wedge \neg C_2 \wedge (\neg D_1 \vee \neg D_0)$

AtMostK via Full Adder Encoding

AtMostK$([x_1, x_2, x_3, x_4], 2)$ Enforce: $x_1 + x_2 + x_3 + x_4 \leq 2$

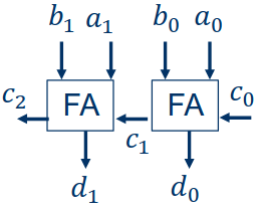
Encoding using two Full Adders

First FA:

$$A_0 = x_1$$
$$B_0 = x_2$$

Second FA:

$$A_1 = x_3$$
$$B_1 = x_4$$



Constraints: $\neg C_0 \ \wedge \ \big(C_2 \implies (\neg D_1 \wedge \neg D_0 \wedge \neg C_1)\big)$